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Machine Learning for
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Assignment 4

As discussed in the lecture, single-cell data with spatial resolution can be used to detect and define cellular niches. Use the provided dataset *data.parquet* from the paper below for the analysis. The columns "X" and "Y" contain the physical coordinates, "sample" represents the sample origin.

Christian M. Schürch et al. *Coordinated Cellular Neighborhoods Orchestrate Antitumoral Immunity at the Colorectal Cancer Invasive Front*, Cell, Volume 182, Issue 5, 2020
https://doi.org/10.1016/j.cell.2020.07.005.

Task 1 KNN-based neighborhood analysis. (60 P)

Redo the neighborhood analysis described in the original publication. Follow these steps:

A. Construct neighborhoods using a KNN-graph with $k = 10$ for each cell. (15 P)

B. For each cell in each neighborhood, calculate the distribution of cell types (column "Celltype"). (25 P)

C. Using these features, cluster the cells using `MiniBatchKMeans` into 10 clusters (15 P)

D. For each cluster, calculate the enrichment score with the given `enrichment_score` function and visualize the result in a suitable manner. (5 P)

Task 2 Radius-based neighborhood analysis (40 P)

An alternative approach to constructing neighborhoods is using a radius-based nearest-neighbor graph. Here, a cell is considered to be in the neighborhood of another cell if it is spatially closer than a predefined radius.

A. Run the analysis from Task 1 again with a radius r. Select r by calculating the median distance of the 10th neighbor $\tilde{x}$ and the corresponding median absolute deviation (MAD). Choose $r = \tilde{x} + \text{MAD}$. (20 P)

B. Compare to the original result in terms of Spearman correlation between enrichment scores and adjusted rand index between clusters. Discuss the results. (10 P)

C. What are the advantages and disadvantages of both ways of constructing neighborhoods? (10 P)

Task 3 (Bonus) Compositionality (20 P)

A. Even though well known and described within statistics (composition statistics), most computational biologists ignore *compositionality* in their data. Explain this term and how it applies in this analysis. (10 P)

B. Redo your analysis while considering the compositionality of features. For simplicity, apply isometric log-ratio transformation and multiplicative replacement as replacement strategy. Use the neighborhoods from Task 2. Again cluster with $k = 10$ and compare to the original result (Task 1) in terms of Spearman correlation between enrichment scores and adjusted rand index. Briefly discuss the differences. (10 P)